1	**(a)**	$F_{\text{on fuel}} = v\dfrac{dm}{dt}$	C1
		$= 3.0 \times 10^4 \times 1.7 \times 10^{-6} = 0.051\text{N} = F_{\text{on rocket}}$	A1
	(b)	Since F is constant and a = F/m, with *m* <u>decreases</u> with time, *a* <u>increases</u> with time.	B1
	(c)	change in velocity = area under graph $= \frac{1}{2}(9.45 + 8.20) \times 10^{-5} \times 4.80 \times 10^{7} = 4240$ m s^{-1}	M2
		final v = 4240 - 0 = 4240 m s^{-1}	A1
	(d)	Shape is parabolic from t = 0 to t = 4.8 x 10^7 s, starting from v = 0 to final v at 4240 m s^{-1}	B1
		Between t = 4.8 x 10^7 to t = 6.0 x 10^7 s, no more force so constant v at 4240 m s^{-1}	B1
2	**(a)**	speed (of object) at surface (of planet) / specified starting point	B1